INCH-POUND

FF-S-210C
07 MAR 2022
SUPERSEDING
FF-S-210B
16 JUL 1981

DETAIL SPECIFICATION SHEET

SETSCREWS: SQUARE HEAD (INCH) AND SLOTTED HEADLESS (INCH AND METRIC)

Inactive for new design after 15 December 2000
and is no longer used, except for replacement purposes.

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

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1. SCOPE

1.1. Scope. This specification covers square head (inch) and slotted headless setscrews (inch and metric). See 6.1 for recommended style usage limitations.

1.2. Classification.

1.2.1. Types. Setscrews shall be of the following types, as specified (see 6.2).

Type I – Square head, inch dimensions
Type II – Slotted headless, inch dimensions
Type III – Slotted headless, Inch dimensions

1.2.2. Styles. Setscrews shall be of the following styles, as specified (see 6.2).

Style 1 – Flat point
Style 2 – Cone point
Style 3 – Oval point
Style 4 – Cup point
Style 5 – Full-dog point
Style 6 – Half-dog point

1.2.3. Size. Setscrews shall be classified by size in accordance with the basic major diameter of the thread (see 6.2)

1.2.4. Length. Setscrews shall be of the length specified in 6.2

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents.

DEPARTMENT OF DEFENSE STANDARDS

QQ-N-281 – Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forging, and Structural and Special Shaped Sections

FED-STD-H28/2 – Screw-Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications

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FED-STD-H28/21 – Screw-Thread Standards For Federal Services, Section 21, Metric, Screw-Threads

MIL-F-495 – Finish, Chemical, Black, for Copper Alloys

MIL-DTL-13924 – Coating, Oxide, Black, for Ferrous Metals

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product

DS51409 – Setscrew, Slotted Headless, Cone Point, Carbon Steel, Cadmium Plated, Metric

DS51410 – Setscrew, Slotted Headless, Cup Point, Carbon Steel, Cadmium Plated, Metric

DS51411 – Setscrew, Slotted Headless, Flat Point, Carbon Steel, Cadmium Plated, Metric

(Copies of this documents are available online at https://quicksearch.dla.mil.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ).

ASQ H1331 – Zero Acceptance Sampling Plans

(Copies of this document are available for purchase online from www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL.

SAE AIR4127 – Steel: Chemical Composition and Hardenability

SAE AMS-H-6875 – Heat Treatment of Steel Raw Materials

SAE AMS-QQ-P-416 – (R) Plating, Cadmium (Electrodeposited)

(Copies of this document are available for purchase online from www.sae.org.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) STANDARD.

ASME B18.6.2 – Square Head Set Screws and Slotted Headless Set Screws (Inch Series)

(Copies of this document are available for purchase online from www.asme.org.)

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ASTM INTERNATIONAL (ASTM)

ASTM E18 – Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test For

ASTM B21/B21M – Standard Specification for Naval Brass Rod, Bar, and Shapes

ASTM B134/B134M – Standard Specification for Brass Wire

ASTM A342/A342M – Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM E1282 – Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

ASTM A380/A380M – Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems

ASTM SI10 – American National Standard for Use of the International System of Units (SI): The Modern Metric System - Revision IEEE/ASTM SI 10-2010

(Copies of these documents are available for purchase online from www.astm.org.)

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIERMENTS

3.1. Material. Recycled and reclaimed materials to the maximum extent practicable. Setscrews shall be made of one of the following materials, as specified (see 6.2).

3.1.1. Low-carbon steel. Low-carbon steel shall be composition1010 (UNS G10100) threw 1020 (UNS G10200) or free machining low-carbon steel, composition 1112, 1113 and leaded steels in accordance with SAE AIR4127.

3.1.2. Medium-carbon steel. Medium-carbon steel shall be composition 1035 (UNS G10350) thru 1045 (UNS G10450) or 1137 threw 1144 in accordance with SAE AIR4127.

3.1.3. Alloy steel. Alloy steel shall be composition 4032 (UNS G40320), 4037 (UNS G40370), 4137 (UNS G41370), 4140 (UNS G41400), 8630 (UNS G86300), 8740 (UNS G87400) or equivalent in accordance with SAE AIR4127, which can be heat treated in accordance with SAE AMS-H-6875 to meet the hardness requirement of 3.3.

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3.1.4. Corrosion-resistant steel.

3.1.4.1. Austenitic. Austenitic corrosion-resistant steel shall be of the SAE 30300 (UNS S3000) Series in accordance with SAE AIR4127 or any other austenitic corrosion-resistant steel developed for cold heading or free machining.

3.1.4.1.1. Magnetic permeability. Austenitic corrosion-resistant steel setscrews shall have a magnetic permeability of 2.0 max (air – 1.0) for a field strength of H = 200 oersteds (16 kA/m) when using a magnetic indicator in accordance with ASTM A342/A342M.

3.1.4.2. Martensitic. Martensitic corrosion-resistant steel shall be steel nos. SAE 51410 (UNS S41000) or SAE51416 (UNS S41600) in accordance with SAE AIR4127.

3.1.5. Brass. Brass shall be in accordance with alloy no. 462 (UNS C46200) of ASTM B21/B21M or alloy nos. 260 (UNS C26000), 270 (UNS C27000) and 274 (UNS C27400) of ASTM B134/B134M.

3.1.6. Nickel-copper alloy. Nickel-copper alloy shall be in accordance with QQ-N-281.

3.2. Protective finish. Unless otherwise specified (see 6.2), setscrews shall be furnished uncoated. When specified (see 6.2), protective finishes shall be as follows.

3.2.1. Cadmium plating. Carbon and alloy steel setscrews shall be cadmium plated in accordance with SAE AMS-QQ-P-416, type II, class 3 (0.0002 in. thick or 5.1 µm thick).

3.2.2. Passivation. Corrosion-resistant steel setscrews shall be passivated in accordance with ASTM A380/A380M.

3.2.3. Black oxide.

3.2.3.1. Brass. Brass setscrews shall be black oxide coated in accordance with MIL-F-495.

3.2.3.2. Corrosion-resistant steel. Corrosion-resistant steel setscrews shall be black oxide coated in accordance with MIL-DTL-13924, class 4.

3.2.4. Hydrogen embrittlement. Cadmium plated alloy steel setscrews shall be subjected to an embrittlement relief treatment in accordance SAE AMS-QQ-P-416 as soon possible after plating to minimize the resulting embrittlement.

3.3. Hardness. Alloy steel setscrews shall be hardened by heat treatment to 45-53HRC. Low and medium carbon steel setscrews shall be hardened by heat treatment to 30-37HRC or shall be surface hardened to not less than 60HR30N. Martensitic corrosion-resistant steel setscrews shall be hardened by heat treatment to 35-45HRC.

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3.3.1. Decarburization. The complete and partial decarburization of the threads of heat-treated setscrews shall not exceed the amounts shown in figure 1 or the amount for any other class as may be specified in the contract to order. A carbon content throughout the threads equivalent to that of the core is desirable.

3.4. Design.

3.4.1. Dimensions. Dimensions and tolerances of setscrews with inch dimensions shall be in accordance with ASME B18.6.2. Slotted setscrews with metric dimensions shall be in accordance with applicable military standards.

3.4.2. Threads. Unless otherwise specified (see 6.2) the inch system threads shall be class 2A in both UNC and UNF series in accordance with FED-STD-H28/2. Metric system threads shall be "M" profile tolerance class 6g, before coating or plating in accordance with FED-STD-H28/21.

3.5. Workmanship. Setscrews shall be free from burrs, seams, laps, loose scale, irregular surfaces, and any other defects which may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Inspection Provisions.

4.2.1. Lot. A lot shall consist of all setscrews of the same type, style, material, protective finish, size and thread produced under essentially the same conditions and offered for acceptance at one time.

4.2.2. Sampling for examination. Unless otherwise specified in the contract (see 6.2), for sampling for examination, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. Acceptable verification levels for MIL-STD-1916 and index values for ASQ H1331 shall be specified in Table I.

4.2.3. Sampling for hardness and decarburization tests. Unless otherwise specified in the contract (see 6.2), for sampling for hardness and decarburization tests, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. A random sample of setscrews shall be taken from each lot in accordance with MIL-STD-1916, verification level IV or ASQ H1331 with an index value of 1.0.

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4.2.4. Sampling for protective finish tests. Sampling for tests of protective finishes shall be in accordance with applicable specifications in 3.2.

4.2.5. Sampling for chemical analysis. When specified (see 6.2), a random sample of setscrews shall be taken from each lot in accordance with MIL-STD-1916, verification level III or ASQ H1331 with an index value of 1.5. Samples previously selected for examination or for the hardness and decarburization teste may be used.

4.2.6. Sampling for examination of packing and packaging. Sampling for examination of packaging, packing and marking shall be specified in the contract order (see 5.1).

4.3. Examination. Each setscrew taken as specified in 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table I. Any setscrew in the sample containing one or more defects shall be rejected and if the number of defective setscrews in any sample exceeds the number of that sample, the lot represented by the sample shall be rejected.

Table I. Classification of Defects

Categories	Defects	Inspection method
Critical	None Define	
Major	Verification level III, Index Value=2.5	
101	Thread	SIE*
102	Width across flats	SIE
Minor	Verification level II, Index value=4.0	
201	Thread length	SIE
202	Applicable point dimensions	SIE
203	Setscrew length	SIE
204	Other dimensions	SIE
205	Eccentricity	SIE
206	Workmanship (see 3.5)	Visual
207	Magnetic permeability(see3.1.4.1.1)	SIE

*SIE = Standard inspection equipment

4.4. Tests.

4.4.1. Hardness. Each setscrew taken as specified in 4.2.3 shall be subjected to hardness test in accordance with ASTM E18.

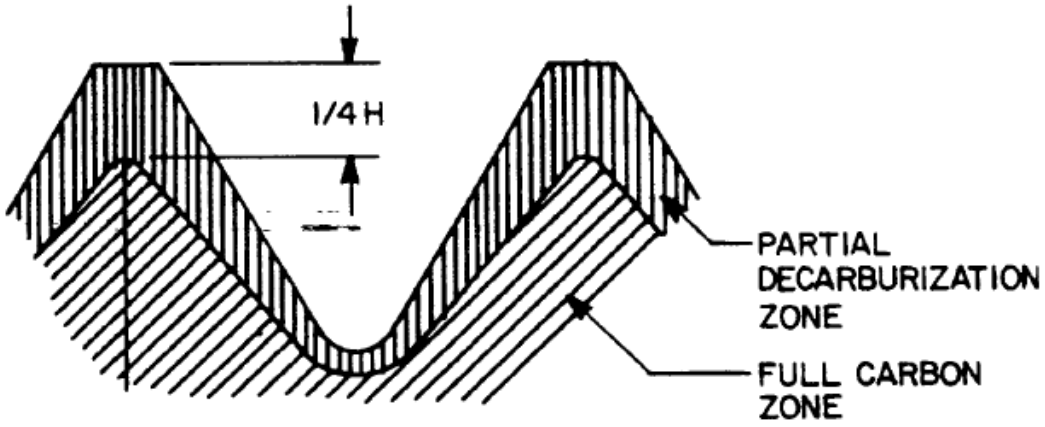
a. Setscrews, having a length of 2 diameters or longer, shall be tested on a transverse section through the threads at a point one-quarter of the nominal diameter from the axis and one nominal diameter from the point.
b. Setscrews having length less than 2 diameters shall be tested on a transverse section through the threads at a point one-quarter of the nominal diameter from the axis and located at the first full thread from the point.

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c. Setscrews having nominal diameters of .25 inches (6mm) and larger shall be tested using the Rockwell C scale. For sizes smaller than .25 inches (6mm) the Rockwell 30N scale shall be used.

4.4.2. Decarburization. Each setscrew taken as specified in 4.2.3 shall be subjected to a decarburization test conducted as follows: The threaded part of the setscrew shall be sectioned longitudinally through the axis and polished so that rounding of the thread edges is held to a minimum. The section shall be etched with the usual metallographic etchants (such as nital) and examined with metallurgical microscope. Measurements shall be made from the crest of the thread to the end of the decarburized zone. This distance shall not exceed that indicated in figure 1 for inch and metric setscrews. Measurements shall be made with a fixed-scale micrometer eyepiece. This test does not apply to austenitic corrosion-resistant steel.

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Inch	
Threads per inch	1/4 H 1/ inches
80	.002
72	.002
64	.003
56	.003
48	.003
44	.004
40	.004
36	.005
32	.005
28	.006
24	.007
20	.008
18	.009
16	.010
14	.012
13	.012
12	.014
11	.015
10	.016
9	.018
8	.020
7	.023
6	.027

1/ 1/4 H = 1/4 BASIC THREAD DEPTH

Metric	
Thread Pitch	1/4 H 1/ mm
0.30	0.047
0.40	0.054
0.45	0.061
0.50	0.068
0.70	0.095
0.80	0.108
1.00	0.135
1.25	0.169
1.50	0.203
1.75	0.237
2.00	0.271
2.50	0.338
3.00	0.406

1/ 1/4 H = 1/4 BASIC THREAD DEPTH

Figure 1. Decarburization limits.

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4.4.3. Protective finish. Protective finish tests shall be conducted in accordance with the applicable specifications in 3.2.

4.4.4. Chemical analysis. Each setscrew selected as specified in 4.2.5 shall be subjected to a chemical analysis conducted in accordance with ASTM E1282 to verify conformance to the applicable material specification. A steel mill certificate will be acceptable in lieu of this test.

4.4.5. Examination and test of packaging and packing. Examination and test of packaging, packing and marking shall be specified in the contract order (see 5.1).

4.4.6. Hydrogen embrittlement. The contractor shall furnish the government certification that cadmium plated alloy steel setscrews have been subjected to the embrittlement relief treatment specified in 3.2.4.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1. Intended use. Setscrews covered by this specification are intended to be used to locate machine parts in relation to each other, either permanently or temporarily. It is recommended that cup point setscrews, style 4, be used wherever possible. For hardened shafts or when the cutting in of the cup point is objectionable, styles 1 or 6 should be used. It is recommended that styles 2, 3, or 5 be used only when it is impactable to use the other styles. For metric setscrews, it is recommended that styles be limited to 1, 2, or 4.

6.2. Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Title, number and date of Department of Defense standard if applicable.
c. Department of Defense standard part number, if applicable.
d. Type, style, size, length (see1.2).
e. Thread series and class (see 3.4.2).
f. Material (see 3.1)
g. Protective finish, if required (see 3.2)
h. Chemical analysis, when required (see 4.2.5).
i. Sampling for examination method, if applicable (see 4.2.2)

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j. Sampling for hardness and decarburization tests method, if applicable (see 4.2.3)
k. Levels of preservation, packaging and packing required (see 5.1).

6.3. Metric terms and definitions. Metric terms used in this specification are defined in ASTM International Standard ASTM SI21, Standard for Metric Practice.

6.4. Military procurement. Items procured under this specification for military use are to be limited (whenever possible) to the variety shown on the applicable Department of Defense standards. Personnel of military departments are requested to refer to these documents for guidance.

6.5. Supersession data. Cross reference between the types and styles of setscrews covered in part by this specification and FF-S-200, and the types and styles of the superseded FF-S-103, dated October 7, 1958, is as follows:

FF-S-103 Types	FF-S-210 Types	FF-S-200 Types
Type I – Square Head	Type I – Square Head – Inch	– – –
Type II – Slotted Headless	Type II – Slotted Headless – Inch	– – –
Type III – Hexagon Socket, Headless	– – –	Type I – Hexagon Socket, Headless
Type IV – Fluted Socket, Headless	– – –	Type II – Spline Socket, Headless
	Type III – Slotted Headless – Metric	
Styles	Styles	Styles
Style 1 – Flat point Style 2 – Cone point Style 3 – Oval point Style 4 – Cup point Style 5 – Full-dog point Style 6 – Half-dog point	Style 1 – Flat point Style 2 – Cone point Style 3 – Oval point Style 4 – Cup point Style 5 – Full-dog point Style 6 – Half-dog point	Style 1 – Flat point Style 2 – Cone point Style 3 – Oval point Style 4 – Cup point Style 5 – Full-dog point Style 6 – Half-dog point

6.6. Subject term (key word) listing. Cadmium

6.7. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a

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